

CONTACT
INFORMATION

School of Computing
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RESEARCH
INTERESTS

Human-Computer Interaction (HCI), Creativity Support, Game Design,
Human-AI Collaboration, Data Visualization

EDUCATION

National University of Singapore, Singapore

Ph.D. in Computer Science Aug. 2026 -
Advisor: Jane L. E

ShanghaiTech University, Shanghai, China

Master of Computer Science Sept. 2022 - Jun. 2025
Advisor: Quan Li

ShanghaiTech University, Shanghai, China

B.Eng. in Computer Science Sept. 2018 - Jun. 2022
Advisor: Quan Li

ACADEMIC
EXPERIENCE

Palette Lab, National University of Singapore, Kent Ridge Campus, Singapore

Research Internship (Advisor: Jane L. E) 2026 - Present
Research focused on creativity support for **game aesthetic customizations**.

ViSeer Lab, ShanghaiTech University, Shanghai, China

Graduate Research Assistant (Advisor: Quan Li) 2022 - 2025
Research focused on *visual analytics systems* for **game algorithms** and *creativity support tools* for **game storytelling**. My work includes:

- Designed and evaluated a visual analytics framework integrating a scalable churn prediction model with two eXplainable AI methods, providing actionable insights to game designers to reduce player churn. Conducted user interviews and a within-subjects study to validate its effectiveness and usability in real-world scenarios.
- Developed and evaluated a visual analytics framework for in-game friend recommendation systems, incorporating methods to balance similarity and diversity. This framework supports fine-tuning recommendation processes based on player feedback and behavior.
- Created and evaluated an in-game mod for automatic clue capture and classification, enabling interactive and creative story interpretation. Conducted a between-subjects study to evaluate the effectiveness and usability of the system.

The HCI Initiative, Hong Kong University of Science and Technology, Hong Kong, China

Graduate Research Assistant (Advisors: Xiaojuan Ma, Jianping Gan) 2023 - 2024
Visual analytics on **South China Sea**.

- Processed 30 days of 3D ocean data, generating vector fields for ocean currents and performing ocean volume modeling to create a realistic environmental dataset for analysis.
- Built an interactive visual analytics system in Unity 3D, integrating Cinema4D for terrain modeling, Unity VFX Graph for ocean currents, custom shaders for volume rendering, and interactions for precise ocean area localization, enabling users to explore and interact with the ocean environment.

User eXperience Center, NetEase, Shanghai, China

Research Internship (Advisor: Wei Wan)

2022 - 2023 (Overlap with ViSeer Lab)

Contributed to the development of visual analytics frameworks for churn prediction and diverse friend recommendation systems, in collaboration with ViSeer Lab.

ViSeer Lab, ShanghaiTech University, Shanghai, China

Undergraduate Research Assistant (Advisor: Quan Li)

2021 - 2022

Visual analytics systems for e-commerce.

PUBLICATIONS

Yuheng Shao*, Junjie Xiong*, Chaoran Wu, **Xiyuan Wang**, Ziyu Zhou, Yang Ouyang, Qinyi Tao, Quan Li. **WordCraft: Scaffolding the Keyword Method for L2 Vocabulary Learning with Multimodal LLMs**. In *ACM CHI Conference on Human Factors in Computing Systems*.

CHI 2026 · Full Paper

Yang Ouyang, Shenghan Gao, Yuheng Shao, Junjie Xiong, **Xiyuan Wang**, Quan Li. **VidSense: Facilitating Incremental Externalization and Knowledge Consolidation via Canvas-Based Interactions in Video Learning**. In *ACM CHI Conference on Human Factors in Computing Systems*.

CHI 2026 · Extended Abstract

Xiyuan Wang, Yifan Cao, Junjie Xiong, Sizhe Chen, Wenxuan Li, Junjie Zhang, Quan Li. **ClueCart: Supporting Game Story Interpretation and Narrative Inference from Fragmented Clues**. In *ACM CHI Conference on Human Factors in Computing Systems*.

CHI 2025 · Full Paper · **Best Paper Award** 🏆

Yuanhao Zhang, Yumeng Wang, **Xiyuan Wang**, Changyang He, Chenliang Huang, Xiaojuan Ma. **CoKnowledge: Supporting Assimilation of Time-synced Collective Knowledge in Online Science Videos**. In *ACM CHI Conference on Human Factors in Computing Systems*.

CHI 2025 · Full Paper

Xiyuan Wang, Ziang Li, Sizhe Chen, Wei Wan, Xingxing Xing, Quan Li. **Prefer2SD: A Human-in-the-Loop Approach to Balancing Similarity and Diversity in In-Game Friend Recommendations**. In *ACM Conference on Intelligent User Interfaces*.

IUI 2025 · Full Paper

Quan Li, Yun Tian, **Xiyuan Wang**, Laixin Xie, Dandan Lin, Lingling Yi, Xiaojuan Ma. **MetapathVis: Inspecting the Effect of Metapath in Heterogeneous Network Embedding via Visual Analytics**. In *Computer Graphics Forum*.

CGF 25 · Full Paper

Laixin Xie, Ying Zhang, **Xiyuan Wang**, Shiyi Liu, Shenghan Gao, Xingxing Xing, Wei Wan, Haipeng Zhang, Quan Li. **Influence Maximization in Temporal Social Networks with a Cold-start Problem: A Supervised Approach**. In *International AAAI Conference on Web and Social Media*.

ICWSM 25 · Full Paper

Yang Ouyang, Yuchen Wu, **Xiyuan Wang**, Laixin Xie, Weicong Cheng, Jianping Gan, Quan Li, Xiaojuan Ma. **OceanVive: An Immersive Visualization System for Communicating Complex Oceanic Phenomena**. In *IEEE Transactions on Visualization and Computer Graphics (Proc. VIS 2025)*.

IEEE VIS 2025 · Short Paper

Xiyuan Wang*, Laixin Xie*, He Wang, Xingxing Xing, Wei Wan, Ziming Wu, Xiaojuan Ma, Quan Li. **Deciphering Explicit and Implicit Features for Reliable, Interpretable, and Actionable User Churn Prediction in Online Video Games**. In *IEEE Transactions on Visualization and Computer Graphics*.

IEEE TVCG 24 · Full Paper

Longfei Chen, Chen Cheng, He Wang, **Xiyuan Wang**, Yun Tian, Xuanwu Yue, Wong Kam-Kwai, Suting Hong, Quan Li. **FMLens: Towards Better Scaffolding the Process of Fund Manager Selection in Fund Investments**. In *IEEE Transactions on Visualization and Computer Graphics*.

IEEE TVCG 24 · Full Paper

Chenyang Zhang, **Xiyuan Wang**, Chuyi Zhao, Yijing Ren, Tianyu Zhang, Zhenhui Peng, Xiaomeng Fan, Xiaojuan Ma, Quan Li. **PromotionLens: Inspecting Promotion Strategies of Online**

E-commerce via Visual Analytics. In *IEEE Transactions on Visualization and Computer Graphics (Proc. VIS 2022)*.
IEEE VIS 2022 · Full Paper

CONFERENCE PRESENTATIONS	ClueCart: Supporting Game Story Interpretation and Narrative Inference from Fragmented Clues. ACM CHI 2025, Yokohama, Japan.	
	A Visual Analytics Approach to Exploring the Feature and Label Space Based on Semi-structured Electronic Medical Records. IEEE VIS 2023, Melbourne, Australia.	
	PromotionLens: Inspecting Promotion Strategies of Online E-commerce via Visual Analytics. ChinaVis 2022, Xining, China.	
HONORS AND AWARDS	Outstanding Graduate by Shanghai Municipal Education Commission	2025
	Outstanding Graduate by ShanghaiTech University	2025
	Best Paper Award by ACM CHI 2025	2025
	Third Price of ChinaVis Data Challenge	2023
	Second Price of ChinaVis Data Challenge	2022
SERVICES	CHI; CHI lbw; Chinese CHI , Conference	
	<i>Peer Review</i>	2022 - 2026
	ARTS 1422 - Data Visualization , ShanghaiTech University, China	
	<i>Teaching Assistant</i>	Spring 2023
	Kedu No. 1 Primary School , Qiannan Buyi and Miao Autonomous Prefecture, China	
	<i>Volunteer Teacher</i>	2019
SKILLS	Shanghai Marathon , Shanghai, China	
	<i>Volunteer</i>	2018, 2020
	Research: Data Visualization, Human-Computer Interaction, Quantitative & Qualitative Research, Interview, Iterative Design.	
	Program Language: JavaScript, Python, C Sharp, SQL.	
	Framework and Tools: D3, Vue, MongoDB, PyTorch, Unity 3D, Cinema 4D, Figma.	
	Languages: Mandarin Chinese (Native), English (IELTS: 6.5, DET: 130), Visualization, L ^A T _E X.	